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Review article

# Adaptive user interfaces and universal usability through plasticity of user interface design

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ABSTRACT

A review of research on universal usability, plasticity of user interface design and facilitation of interface development with universal usability is presented. The survey was based on 165 research papers spanning over fifty-five years. The foundations of adaptive or intelligent user interfaces (AUI or IUI) are presented, three core domains being focused upon: Artificial Intelligence (AI), User Modelling (UM) and Human–Computer Interaction (HCI). For comparison of the various AUIs, a proposed taxonomy is given. One conclusion is that an efficient training vector for fast optimal convergence of the machine-learning algorithm is a necessity, but key to this is the bounding of the dataset, the goal being to achieve an accurate user preference model, which has to be built from a limited number of datasets obtained from the human interaction. More research also needs to be conducted to ascertain the usefulness and effectiveness of IUIs compared against AUIs.

With the global mobility of users, interface design must take account of the abilities and cultures of users, derived from actual user behaviour and not on their feedback. A key question is whether the interface should be adaptive under system control or be made adaptable under user control. A need is identified for an "afferential component" that stores a priori information about the end user, an "inferential component" that determines to what extent the user interface actually needs to be adapted, and the "efferential component" that actually determines how the adaptivity is applied seamlessly to the system. Application to e-learning is a priority: the use of machine intelligence to achieve appropriate learnability, ideally enhanced by "Playful interaction", was found to be desirable. Universal application of adaptation lies in the future, but AUI properties cannot be ascertained while disregarding the other parameters of the system in which it will be used. A more complete understanding of the human mental model is necessary, requiring a highly multidisciplinary approach and cooperation between diverse researchers.

Finally, a performance evaluation of plasticity of user interface was conducted: it is concluded that the use of dynamic techniques can enhance the user experience to a much greater extent than more basic approaches, although optimisation of usability parameter trade-offs needs further attention.

It is noted that most of the work reviewed originated from a limited range of cultural perspectives. To make an interface simultaneously usable for users from a diverse range of cultural backgrounds will require a very large amount of adaptation, but the powerful principles of plasticity of user interface design hold the future promise of an optimum tool to achieve cross-cultural usability.



**Contents**



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## 1. Introduction

The goal of this review is to summarise the literature on implementation of universal usability using dynamic user interface design techniques and to draw appropriate conclusions. A representative sample of research is presented, spanning over 55 years, starting from the early work in this field. Different types of users, interfaces, applications, technologies and methods were investigated. Identifying the most propitious direction for future adaptive user interface design is the main focus of the review study.

Computer interfaces are becoming more complex to cater for more features whilst at the same time having to present a simpler interface to the user. This is in tandem with the accrescent complexity of software systems and the networked information systems. These increases in complexity are necessary to cater for the widening and growing diversity of human users interacting with the systems. Many of these users have never interacted with a computer system before, such as the elderly and non-technical professional people. The need to provide accessibility to disabled, handicapped and sensory-deprived people is also increasing. The intellectual, skill, health and emotional statuses of the user also need to be accommodated. The initial approach was to cater to the needs of the absolute beginner; however, recent developers are now providing 'plasticity' of user interfaces to mould to the needs of the user's past experiences and actual current needs and requirements.

The research presented here was based on three central themes:

(i) the establishment of the history and the current status of the universal usability of user interface design;
(ii) reviewing the historical progress and the current status of plasticity of user interface design and
(iii) the facilitation of development of universal user interfaces by deploying adaptation mechanisms.

To address all these issues, an extensive literature survey has been conducted.

The knowledge domain of the research is in the realm of human–computer interaction, specifically, in ubiquitous or pervasive computing [1,2], user interface design [3], usability [4] and plasticity of user interface design [5].

## 2. Methodology

An extensive literature search has been conducted, taking the line of Barbara Kitchenham and Stuart Charters [6] of the middle path between Systematic Literature Review (SLR) and Secondary Study. Most of the features of an SLR were taken in this paper. The particular definitions of these terms according to [6] are reproduced below:

> "Systematic Literature Review (also referred to as a Systematic Review): A form of secondary study that uses a well-defined methodology to identify, analyse and interpret all available evidence related to a specific research question in a way that is unbiased and (to a degree) repeatable". [6]

"Secondary Study: A study that reviews all the primary studies relating to a specific research question with the aim of integrating/synthesising evidence related to a specific research question". [6]

The results obtained have been organised in a mixed mode: thematically and chronologically within a subject field. The whole process involved the following two stages:

### 2.1. Search process

Using keywords in a literature-searching programmes, a comprehensive search was conducted using the following databases: ACM Digital Library, EBSCO Host, IEEE Xplore, Wiley, Web of Knowledge (ISI), Springer, Science Direct, Scopus, Sage Journals, Oxford Journals, Cambridge Journals, Saudi Digital Library (SDL) and SciSearch.

The search strings were generated by utilising different permutation and combination of the keywords such as Dynamic UI Design, Plasticity, Adaptive User Interface (AUI), Adaptation, Adaptability, Adaptivity, Universal Usability, Inclusivity and Inclusive Design. For instance, one of the search strings used in IEEE Xplore is "((((((((((("All Metadata":Dynamic UI Design) OR "All Metadata":Plasticity) OR "All Metadata":Adaptive) OR "All Metadata":Adaptation) OR "All Metadata":Adaptability) OR "All Metadata":Adaptivity) OR "All Metadata":Universal Usability) OR "All Metadata":ubiquitous) OR "All Metadata":Inclusive Design) OR "All Metadata":pervasive) AND "All Metadata":User Interface Design)". A well-defined inclusion and exclusion criteria, as set in Section 2.2, was also applied, along with the key terms, in order to limit the search result within the focus of the study. The study covers a duration of over 55 years, starting from 1960 to 2015. The literature search was last updated in early 2018 and the draft of the article was completed as well as submitted to the journal in late 2019.

In fact, SDL is a subscription based integrated service that comprises many literature databases, including most of the above-mentioned ones. Therefore, the search produced some duplicate results, which were later eliminated. Table 1 displays the number of articles retrieved through the initial search and finally selected after applying the inclusion and exclusion criteria.

### 2.2. Study inclusion and exclusion criteria

The approach adopted was to focus on all database studies that appeared in peer-reviewed journals and in highly regarded conferences in the English language. Dissertations, theses, or presentation abstracts that were not published in peer-reviewed journals and conference proceedings were excluded. The initial search identified over 300 studies, but only 165 met the inclusion criteria and were included in the review. The Critical Appraisal Tool as suggested by Wallace and Wray (13 January 2011) [7] has been used. This tool consists of a list of questions that need to be asked while evaluating any literature for research. It is recommended that the literature be included only if the answers to those questions seem to be satisfactory for the researchers.

Year-wise list of articles that meet the inclusion and exclusion criteria is provided in Table 2, to demonstrate how the researchers' degree of interest on this topic has increased.

## 3. Background terminology

### 3.1. Human–Computer Interaction (HCI)

A lack of consensus regarding what constitutes the spectrum of cognate subjects was the major debate regarding the ACM SIGCHI (Special Interest Group on Computer–Human Interaction) Curricula for HCI [135]. However, a practical definition was recognised in order to permit implementation, before moving on to agreement of a definition for the development of educational resources in HCI. The practical definition agreed upon was: "Human–computer interaction is a discipline concerned with the design, evaluation and implementation of interactive computing systems for human use and with the study of major phenomena surrounding them" [135].

Sears and Jacko [129] rightly point out the difficulty of intercomparison of the HCI literature due to the non-existence of an agreed standard definition for HCI. Various alternative terminologies and acronyms in common usage include:

– CHI (Computer–Human Interaction): this term was historically mainly associated with Computer Science though not anymore. It is to be noted that a restricted focus, has been preferred by the ACM Special Interest Group "SIGCHI";
– HF: Human Factors;
– E: Ergonomics, prevalent in Europe.

Researchers, however, use both the terms Human Factors and Ergonomics interchangeably.

### 3.2. User interface

The User Interface is the means by which the user directly engages with the device. Making this interface complicated adversely affects this engagement. However, progress in technology allows more flexibility although it adds complexity in both use and design of the interface [127].

The successful design of the interface in terms of its usefulness depends on several parameters, such as: the mode of interaction; the intention and overt motivation of the users themselves; the learning curve necessary; the cultural background of the users and the technology used to service the interface [4].

### 3.3. User Interface Design (UID)

Muhlhauser and Gurevych [127] allude to the fact that designing and constructing a good user interface is actually much like a "lost art", which has been neglected in the ICT domain.

Two approaches may be taken in designing the user interface: either it can be made intelligently adaptive, or separate tailor-made interfaces may be created. The first type is theoretically responsive to what the user wants. The second type, however, adds different complexity for the designer as the clear delineation of components of the interface becomes increasingly difficult, especially in the case of ubiquitous computing [127]. As computers become more embedded, tending to invisibility through miniaturisation, this may cause problems for the elderly when trying to interact with them. However, this is not a problem so much for those accustomed to growing up with such technology. As embedded and hidden computers permeate our society, interacting with them will become more and more a routine part of everyday life.

### 3.4. Usability

Usability engineering is an iterative process; however, no standardised or agreed procedure exists that ensures complete success and satisfaction. The most important issues have been summarised by Dix et al. [85], Neilsen [29] and Shneiderman and Plaisant [3].

ISO 9241-110 [165] and ISO 9241-400 [166] define usability as, "the extent to which a product can be used by specified end-users to achieve specified goals with effectiveness, efficiency and satisfaction in a specified context of use".

**Table 1**
Articles per database.

| Database | Articles (Initial selection) | Articles (Final selection) |
|---|---|---|
| ACM Digital Library | 103 | 57 |
| IEEE Xplore | 51 | 27 |
| Springer Link | 68 | 33 |
| Science Direct | 37 | 20 |
| Saudi Digital Library (SDL) | 47 | 28 |
| Total | 306 | 165 |

**Table 2**
Year-wise List of Articles.

| Year | Number of articles | Articles |
|---|---|---|
| 1960 | 1 | Licklider [8] |
| 1971 | 1 | Lucas [9] |
| 1973 | 1 | Mason and Mitroff [10] |
| 1977 | 1 | Feeney and Hood [11] |
| 1982 | 1 | Pawlak [12]. |
| 1988 | 2 | Hancock and Chignell [13]; Lai et al. [14] |
| 1989 | 3 | Finin [15]; Mitchell and Shneiderman [16]; Norcio and Stanley [17]. |
| 1990 | 3 | Browne et al. [18]; Crimi et al. [19];MacLean et al. [20]. |
| 1991 | 3 | Arcieri et al. [21]; Robertson et al. [22]; Zimek [23]. |
| 1993 | 9 | Benyon and Murray [24]; Browne [25]; Cote-Munoz [26]; Dieterich et al. [27]; Koller [28]; Neilsen [29]; Sherman and Shortliffe [30]; Schneider-Hufschmidt et al. [31]; Sukaviriya and Foley [32]. |
| 1994 | 5 | Brajnik and Tasso [33]; Carolis and Rosis [34]; McNeill and Thro [35]; Oppermann [36]; Oppermann [37]. |
| 1995 | 7 | Arcand and Ramstein [38]; Hayes-Roth et al. [39]; Kay [40]; Kobsa and Pohl [41]; Okada [42]; Orwant [43]; Savidis and Stephanidis [44]. |
| 1996 | 4 | Bergman et al. [45]; Brickman et al. [46]; Brusilovsky [47]; Ferrucci et al. [48]. |
| 1997 | 3 | Akoumianakis and Stephanidis [49]; Averbukh et al. [50]; Perry et al. [51]. |
| 1998 | 3 | Horvitz et al. [52]; Savidis and Stephanidis [53]; Stephanidis et al. [54]. |
| 1999 | 3 | Langley [55]; Moray, August [56]; Thevenin and Coutaz [57]. |
| 2000 | 13 | Akoumianakis and Stephanidis [58]; Emiliani [59]; Emiliani and Stephanidis [60]; Newell and Gregor [61]; Perkowitz and Etzioni [62]; Ritter et al. [63]; Savidis, Akoumianakis and Stephanidis [58]; Shneiderman [64]; Stephanidis [65]; Stephanidis [66]; Stephanidis et al. [67]; Vanderheiden [68]; Vanderheiden [69]. |
| 2001 | 7 | Brusilovsky [70]; Calvary et al. [71]; Reichenbacher [72]; Schneider and Cordy [73]; Stephanidis [74]; Stephanidis and Savidis [75]; Ritter and Young [76]. |
| 2002 | 5 | Billsus et al. [77]; Brown et al. [78]; Calvary et al. [79]; Newell and Gregor [80]; Shneiderman [81]. |
| 2003 | 8 | Calvary et al. [82]; Daâssi et al. [83]; Demeure and Calvary [84]; Dix et al. [85]; Grundy and Yang [86]; Nyongesa et al. [87] Oviatt [88]; Pieper et al. [89]. |
| 2004 | 11 | Balme et al. [90]; Bunt et al. [91]; Calvary et al. [92]; Findlater and McGrenere [93]; Gajos and Weld [94]; Kerievsky [95]; Mens and Tourwé [96] ); Mori et al. [97]; Oviatt and Seneff [98]; Oviatt et al. [99]; Paymans et al. [100]. |
| 2005 | 7 | Byrne [101]; Grammenosa et al. [102]; Gregor et al. [103]; Limbourg et al. [104]; Savidis et al. [105]; Sottet et al. [5]; Thevenin et al. [106]. |
| 2006 | 7 | Clerckx et al. [107]; Duarte and Carriço [108]; Feng et al. [109]; Muda and Mohamed [110]; Prammanee et al. [111]; Takeuchi and Sugimoto [112]; Perugini and Ramakrishnan [113]. |
| 2007 | 9 | Alvarez-Cortes et al. [114]; Coutaz et al. [115]; Granić and Nakić [116]; He and Yen [117]; Kitchenham and Charters [6]; McGrenere et al. [118]; Sottet et al. [119]; Uflacker and Busse [120]; Wakkary and Hatala [121] |

(*continued on next page*)

ISO 9241-400 [166] also lists the following five important design components of usability engineering:

(i) **Safety and Security** the interface must not be harmful and ought to help to reduce the user error;

(ii) **Effectiveness**: the interface should aid and not hinder the task of the user;

(iii) **Efficiency and Functionality**: the interface should expedite the user task;

**Table 2** (*continued*).

| Year | Number of articles | Articles |
|---|---|---|
| 2008 | 11 | Clerckx et al. [122]; Findlater and McGrenere [123]; Gajos et al. [124]; He et al. [125]; Jameson [126]; Kortum [2]; Muhlhauser and Gurevych [127]; Newell [128]; Rubin and Chisnell [4]; Sears and Jacko [129]; Vanderdonckt and Calleros [130]. |
| 2009 | 11 | Blumendorf et al. [131]; Calvary and Demeure [132]; Grammenos et al. [133]; Grammenos et al. [134]; Hewett et al. [135]; Jayapandian and Jagadish [136]; Reinecke and Bernstein [137]; Sendín and López [138]; Shneiderman and Plaisant [3]; Shi-wei and Shou-Qian [139]; Takeuchi and Sugimoto [140]. |
| 2010 | 11 | Balaram [141]; Blumendorf et al. [142]; Coutaz [143]; Gajos et al. [144]; Jason et al. [145]; Langdon et al. [146]; Lavie and Meyer [147]; Rudin-Brown [148]; Savidis and Stephanidis [149]; Serna et al. [150]; Sánchez et al. [151]. |
| 2011 | 7 | Burzagli et al. [152]; Ceret [153]; Hou et al. [154]; Miraz et al. [155,155]; Reinecke and Bernstein [156]; Tchankue et al. [157]. |
| 2012 | 2 | Mejía et al. [158]; Rogers et al. [1]. |
| 2013 | 4 | Akiki et al. [159]; Akiki [160]; Akiki et al. [161]; Miraz et al. [162]. |
| 2014 | 1 | Miraz et al. [163] |
| 2015 | 1 | Oliveira et al. [164] |

(iv) **Joy and Fun**: use of the interface should not be a laborious endeavour, and
(v) **Ease of Learning and Memorising**: the learning curve should not be steep, and the procedures learned should be memorable.

The above list is not exhaustive and trade-offs are necessary, as well as efforts to reduce conflicting demands.

Usability effectiveness should have either a quantitative or qualitative metric [127]. User Centred Design (UCD), as discussed in Section 3.3 explains that the user involvement is by nature an iterative process. The key is to strike a balance between the various demands of an interface, consisting of: the amount of interaction required; the nature of the user; the content being delivered and processed; the device used; the environment the interface is being applied under; the technology and the performance constraints of the platform under which the user interface is operating.

### 3.5. Inclusive design/universal design/universal usability

The lack of access to technology for those from deprived socio-economic backgrounds, who tend to lack the connectedness of the affluent, is popularly known as the "digital divide". This gap or "digital exclusion" is particularly acute in developing countries. Consequently, developing countries lag greatly in comparison with the developed countries in deploying technologies [155] notably for those aged 50 and over, and especially for disabled users.

The inequity of the digital divide has been and continues to be addressed in terms of equity, business models, accessibility and outreach markets. The approach of "Inclusive Design" has been propounded in terms of "Universal Usability" [64], "Universal Design" [45,141] and "Design for All". Universal Design can be defined as making an object or service that can be easily utilised by as many people as possible. Human factors and ergonomics are the foundation of inclusive design.

### 3.6. Plasticity or adaptation of user interface design

Based on the literature related to plasticity [5,57,71,73,79,83,84,92,115,119,130,132,138,143,150,151,153,164], it is defined as following the properties, attributes and characteristics of plastic materials themselves when applied to a user interface. This implies that the computer user interface adapts, conforms and moulds itself to the needs of the end user while showing resilience to the changing needs of a variety of end users. The plasticity of the user interface must thus not be fragile (nor brittle) and must always return to its initial default state once again upon completion of the current user's session. This elastic property of the user interface is required so that it can be adaptive to the needs of the next end user, who may be different or may even be the same person again. This same person, due to the previous interaction with the interface, now has experience and a changed biophysical state, so may necessitate a different interface, however subtle.

Plasticity, also commonly known as Adaptation, is defined in terms of how the software can accommodate the particular current physical and mental abilities of the user as well as the situation of use and platform capabilities. The concept of this plasticity or adaptation of user interface design, as described in [74], can be considered as a multi-faceted process involving Source of Adaptation Knowledge, Level of Interaction and Type of Knowledge.

## 4. Universal usability

Universal Usability has so far been enriched by contributions from many leading researchers [146]. The seminal research conducted by the pioneer researchers in the field of Universal Usability and Assistive Technology has been extensively reviewed. This includes the works of key researchers such as: Constantine Stephanidis [65] in Greece, Alan Newell [61,80] in the United Kingdom, Gregg Vanderheiden [68,69] and Neil Scott [51] both from North America. The initial survey focused on disabled users; however, this was later extended to cover the elderly and young users, and those with constrained technologies, such as users with small device screens, no screens and slow network connections [81]. The new literature survey presented here is novel, as far as is known to the authors, in including a focus on cultural aspects of universal usability.

As far back as 1993, Newell [167] was a pioneer in recognising the need to cater to the full spectrum of users, especially the disabled and ageing demographics, in designing an effective human–computer interface.

In a later paper, Newell and Gregor [80] sees "enormous opportunities for the human–computer interface design community" due to the "significant changes in the social, legal, demographic, and economic landscape over the past 10–15 years". However, due to continued mass global migrations and the opening-up of the world economy, it has now become pertinent to address the cultural issues as well.

Newell makes an important observation that it is very "dangerous" to design an interface that caters only for two groups of people, that is, the "ordinary and extraordinary" users. This is because human abilities cannot be categorised in a binary way, since humans have abilities which cover a continuum of functionalities. He explained this reasoning succinctly as "it is more accurate to place people as points within a multidimensional space of individual needs, whose axes represent physical and mental characteristics" [167]. He called it the "human-need continuum" [167]. He thus introduced a new design paradigm of "Design for Dynamic Diversity" to address this continuum. "User Sensitive Inclusive Design" as a methodology to assist its achievement has also been proposed (Gregor et al., 8–10 July, [168]). The term "Sensitive" here replaces the term "Centered" of the traditional "User Centered Design Principles" to "underline the extra levels of difficulty involved when the range of functionality and characteristics of the user groups can be so great that it is impossible in any meaningful way to produce a small representative sample of the user group, nor to design a product which truly is accessible by all potential users" [128]. Newell's identification of the need to cater to this human-need continuum was revolutionary. It may be the case that the needs of the special users are often so unique that only a customised usability interface will suffice.

Even though much progress has been made in addressing the deficiency in usability issues, the important factors identified by Gregor et al. [103] still exist. These are:

(i) there is still much software which does not take into consideration the human usability factors for diverse users;
(ii) some of this software is written by people who are young and highly IT-literate and they thus design software which only caters to their own requirements.

Additionally, a digital divide was identified by Pieper et al. [89] : "the divide between those groups of people who benefit from Information Technology and those who do not or cannot access it": sadly, this divide has still not been completely closed. It is thus of even greater urgency to implement a global software strategy that incorporates user-sensitive inclusive design to help to insert a more precise user requirement in the software design life cycle. Taking this approach should alleviate the divergence between catering to diverse users and what the software offers in practice. However, it is recognised that more research needs to be carried out to address the digital divide and to perhaps retrain highly IT-literate software developers to listen carefully to the requirements of users with special needs.

With the ever increasing and diversifying range of input and interaction devices, an even broader range of people is embraced that the software developers have to cater to and this complicates the design parameters of the human–computer interface even further. Consequently, Stephanidis and Savidis [75] identified three important issues that need to be considered, these being:

(i) the nature and ability of the user in terms of any inherent disability;
(ii) what is the action that needs to be performed; and
(iii) how the task is tied to the nature of the environment, i.e. whether it is for social or business use. However, no mention was made of any aspects of cross-cultural usability.

The User Interface design phase is usually the most rigorous part of the software development lifecycle. The metric for defining software quality has shifted in the last twenty years from being based on such parameters as operational reliability and robustness to a focus on usability. More emphasis is now placed on the universal applicability of the software for all situations and all times [66]. The current trend in HCI is a greater emphasis on individualisation of interactivity, especially taking into account the ability of the user with the software end-product, and dispensing with the need to incorporate any "special features" [75].

## 5. Plasticity of user interface design

The key to accessibility for the ever broadening diversity of users in the HCI field is in fact adaptation, as agreed upon by the leading researchers in this field [18,31,36,39,47,52,75]. Stephanidis and Savidis [75] in fact extend this concept to include automatic adaptation. Adaptation is defined in terms of how the software can accommodate the particular physical and mental abilities of the user as well as the situation of use and platform capabilities [74]. The concept of this plasticity or adaptation of user interface design, as described in [74], can be summarised as adaptation balancing the three different demands of:

(i) the type of information to be displayed;
(ii) the level of interaction and
(iii) the source of knowledge adaptation.

Regarding the innovative suggestion of adopting a dynamic or adaptive user interface design to help people with accessibility issues, the literature survey shows two diametrically opposing views. One view suggests adopting a changing interface [80] while another favours a constant or unchanging interface [81]. The second school of thought, however, is of the opinion that this approach might be effectively used "for content in news sites and recommendations on e-commerce sites only". The literature from both camps did not clarify what their sample sizes were. The authors of the present paper proffer a middle viewpoint in that the users should be given the choice to retain or discard certain suggested interface options with an easy choice to go back to the default mode. This, however, needs to be researched in detail.

### 5.1. Types of plasticity or adaptation

How the interface can actually change or mould to the user's needs may be seen in the realm of different factors, such as: the time of the incident of the adaptation; the agents present that are involved in the controlling of the interface and the actual level and depth of the adaptation itself that needs to be carried out [27]. Interactive IT systems may be categorised into either an adaptable system or an adaptive system or both as Mixed-Initiative.

#### 5.1.1. Adaptability

An adaptable system makes use of tools that facilitate the end user to alter the system's characteristics [37]. It is naturally intimately tied to the user's personality and characteristics, though little work exists that truly adapts to the user's personality. This is usually known for the user and is also assumed to remain static during the interaction with the computer. This makes it easier to derive the rules necessary to carry out the adaptation for the user during the first run of the system. The necessary decisions obtained from the rules can then be directly used for the "instantiation of the interaction dialogues" [54]. The steps to carry this out are shown in Fig. 1.

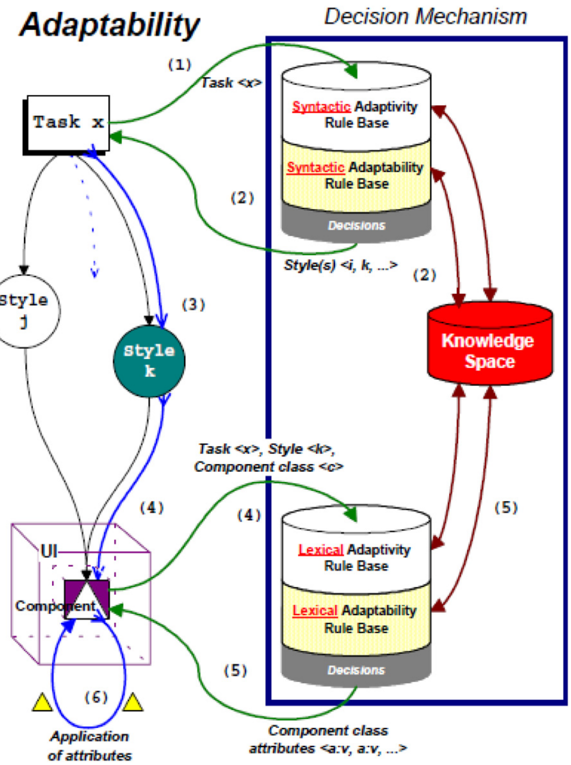



**Fig. 1.** Adaptation (via adaptability) Mechanism used in the AVANTI Project [54].

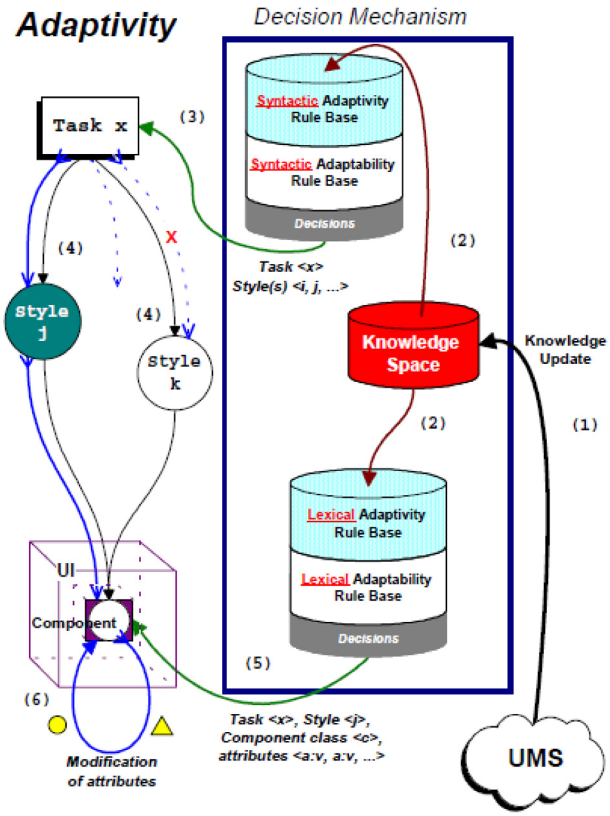



**Fig. 2.** Adaptivity Mechanism used in the AVANTI Project [54].

### 5.1.2. Adaptivity

Adaptivity is only of concern during the runtime. To clarify, the code written during the design phase is known as the design time code. This can be distinguished from when the code is executed, when it is known as the runtime code. It is not started nor triggered by the interface components, as the user characteristics and situation, are not known by them. An adaptive system thus automatically alters its characteristics at runtime, based on assumptions about the user's current runtime usage [27,37]. The adaptations are thus initiated by the decision mechanism. These steps are shown in Fig. 2.

Adaptive User Interfaces (AUI) have been a central component of HCI since its infancy and have always been regarded as a primary characteristic of Intelligent User Interfaces, IUIs [16].

The AUI is a software component, an "artefact", which can dynamically and automatically alter its characteristics in terms of its functionality and interface so as to enhance its effectiveness in interacting with the user by creating a new user model based on some fractional prior experience with that particular user [24,55]. An AUI's *raison d'être* is not to exist in isolation from the user but to only come into existence in order to fully participate with a designated user and only that user. The implication here is that the true adaptive interfaces must be participating with the user throughout their entire session.

It must be stated that the dynamic adaptation of the user interface must only take place if it enhances the way the human user interacts with the computer. If changing the user interface degrades the user's interactivity, then the former User Interface (UI) must be maintained and no change should take place.

The dynamics of adapting the UI are not just a simple matter of remembering or storing the previous interaction of the user. This is because that user's behaviour is also non-constant in terms of how healthy, tired or eager they are when he/she is undertaking a new computer session.

However, the past experience should not be discarded in its entirety as it forms a helpful basis to create a specific AUI for that particular user for his/her new interaction [55].

The make-up, performance and limits of AUIs comprise three primary parts [37] :

***Adaptivity Afferential Component*** the AUI must contain at least some user-related data before it can commence the task of personalising the interface. Individual user information is obtained by the adaptivity afferential component [37]. This user information may be obtained either explicitly or implicitly [114].

***Adaptivity Efferential Component:*** AUIs require automation and must appear totally seamless to the user. The responsibility for this is handled by the efferential component, which dictates both the type and how the adaption is applied to the system [37]. The application of this can be broken down into four levels: information, presentation, user interface and functionality [72, 126].

***Adaptivity Inferential Component****:* To identify the possible need for adaptation, user data is continuously extracted and monitored during the continuing interaction. As this may be considered the vital activity for triggering adaptation, formalised rules must be created for this inference. Thus, two different types of components need to be recorded, one being the type of data that needs to be recorded, the afferential component; the other how the adaption needs to be carried out, the efferential component. The AUI is thus triggered by the type of data of the inferential component [37].

### 5.1.3. Mixed-initiative

Mixed plasticity covers a combination of both human and system intervention [82].

Plasticity or adaptation may be performed automatically from specifications, or manually by human expert designers, depending on the tools available, or by a combination of both with a mixed-initiative locus of control [79].

### 5.2. Issues related to adaptation

With the preceding concerted research efforts in user interface adaptation, there still exist many disagreements amongst the researchers regarding the:

(i) Persistent lack of an internationally agreed definition of what actually are the constituents of user interface adaptation;
(ii) Unified agreement on the software architecture to support the creation of the adaptation system throughout the entire software development lifecycle;
(iii) Type of knowledge components required for reasoning in the various adaptation constituents; and
(iv) Agreement on the threshold values for the adaptation constituents.

In Bunt et al. [91], many differences are specified between adaptive UIs and adaptable UIs. Findlater and McGrenere [93] studied this important relationship between adaptable and adaptive systems.

### 5.3. The journey towards adaptation or plasticity

The first step to this path of adaptability started with the concept of a dual user interface. This integrative approach was designed to cater for blind and sighted users working together, possibly in the same environment [44,53]. The novelty of this idea was the actual inclusion of these considerations right from the beginning of the software design life cycle. Taking this new approach can be considered evolutionary as it led to the basis of incorporating factors to broaden the accessibility beyond these two types of users.

The concept of the dual user interface design was the foundation of the extension of the user interface usability to the concept of "User Interfaces for All".

The literature searches suggest that the "AVANTI" system [54, 67] was the first to utilise adaptive technology to achieve its goal of a user interface that was truly "for all". It was a web-based information system that utilised a conceptual framework to provide both a changing user interface level and a varying content level matched to the user's needs.

The AVANTI system's user interface was designed along the lines of a web browser. This web browser was dynamic as it changed its behaviour from the initial start-up session and then during its session with the user, according to the ability and needs of the user. Thus, the AVANTI system not only dynamically adapted throughout the user's interaction with it, it also accommodated for different individual users as well.

The automation of the user interface adaptation to create the "Unified User Interface" [58]; Savidis, [58] was the next step in the goal towards universal usability. The work known as "USE-IT System" made use of the HAWK toolkit [58]. The approach consisted of identifying all possible scenarios for the various types of users that impacted the software design of the user interface. These were then fed into an abstract model for each of these alternative steps for coding. Finally, the design space was rationalised to accommodate these alternative scenarios [75]. The adaptation rules were generated from the model of the user, the tasks that needed to be performed and the platform on which these operations would be executed, taking into account the restrictions of the device. To facilitate this process, a data structure known as the adaptability model tree was adopted [49].

The research of Grammenosa et al. [102] was pioneering as it integrated the use of "Virtual Prints (ViPs)" in a Virtual Environment (VE). They observed that how a person interacts in a virtual environment needs to be tracked and this can be achieved by leaving markers or signposts with various types of metadata attached. The concept is similar to that of using cookies.

Three types of ViPs may be classified, these being:

(i) Virtual Footprints: for user navigation tracking such as movement, position and orientation;
(ii) Virtual Handprints: for keeping account of how the interaction with the VE had taken place and
(iii) Virtual Markers: created specifically by the user with additional information or metadata, much like an annotated index marker.

An emulation of the ViPs mechanism must be present in order to process and make sense both of the user's ViPs and those of others. The research has shown that a well-designed ViPs mechanism does greatly enhance the usability of VEs [102].

A user study was conducted in two stages. The first stage consisted of disseminating the ideas behind the concept of ViPs to the users to give useful feedback. This feedback was used to refine the ViPs in terms of their usability and value. The second part of the strategy utilised expert assessors to minutely inspect the prototype ViPs in service in the field [102]. It should be noted, however, that the sample size in their first iteration consisted of only six people and in the second iteration consisted of an even smaller sample of five persons.

One of the earliest examples of a dual user interface was the "Starlight" software development platform [44,134]. This was used for interacting with and authoring electronic textbooks that catered for both blind and sighted, including partially sighted, users. The Starlight system was quite innovative in its approach as it offered an advanced form of adaptive prompting with audio cues. This also included offering a selection of shortcuts that were anticipated to be likely to be required by the user or developer.

The dynamic command toolbar provided a context-sensitive direct access to the most popular and heavily used functions and this formed a major component of the main interface. The other major interface component was the dynamic context-sensitive content editor. This was used for the main editing and it also came with a dynamic toolbar [134].

Grammenos et al. [133] also promoted and introduced universal accessibility for handicapped gamers, which was timely, as computer gaming is an established and highly popular form of entertainment. The requirements in terms of hardware, software and user interactivity are quite demanding and these currently exclude a large segment of people with disabilities. To address this limitation the concept of "universally accessible games" was adopted that required minimal or no adjustments to be made by the disabled users. Four case studies based on four different types of games were tested for "key design and evaluation findings". The feedback was collated and subsequently led to the novel solution of "Parallel Game Universes". To explore this, a multi-player game that was accessible to a diverse class of disabled people was developed using Action Script 2.0. People with motor neurone disease, partial sightedness, limited use of limbs and memory problems could all now play this game, known as UA-Chess. Having two different profiles that could switch automatically allowed two different types of people (one able and the other disabled) to now utilise the "two-player mode" to play games using the same hardware. The "universal access objective" was adhered to by the game interface being able to switch between "alternative input and output modalities and interaction methods" [133].

Another early researcher, Pier Luigi Emiliani, conducted extensive research with the campaign "Towards an Information Society for All" [59] and suggested possible adaptation to achieve this aim [60]. He added the ability of the end user to intervene in the process of the adaptation by "the possibility both to insert new data and to alter the structure of ontological knowledge bases which contain adaptation determinants". This was based on Web 2.0 and the Semantic Web. Using this approach was envisioned to offer a more personalised service tailored to the end user's sphere of interest and to keep track of its evolution. [152].

Thus, the foundation work of the early researchers in adaptation could be seen as helping to establish the User Interface Software and Technology (UIST) field to progress towards achieving better and higher quality in implementing interactive systems. This has led to interactive systems that can better match the intentions, interests, preferences and abilities of the user [49].

### *5.4. Some early projects adopting plasticity of user interface design*

Apart from the previously mentioned projects, some other earlier projects in adaptable and adaptive systems include: BUTTONS [20], Xbuttons [22] and OBJECTLENS [14]. The issue of adaptability has also been thoroughly investigated in a European Commission project known as AURA [28].

The principal component of the PODIUM system [30] was the inclusion of adaptability. Other systems also pursued the inclusion of adaptivity in the user interfaces such OPADE [34], AIDA [26] and UIDE [32], including national and international projects like AID [25] and FRIEND21 [42]. Both concepts of adaptability and adaptivity were combined in a single project, as described by Zimek [23]. This consisted of four functional units: user modelling component, task modelling component, strategy component and finally a 'User Interface Management System' (UIMS) [23,44,53]. In addition, Arcieri et al. [21] put forward an alternative architecture that achieved a similar effect.

In addition to the above architectures for adaptable and adaptive user interfaces, there have been a few other proposals, which, however, are narrower in scope. Examples of some early projects supporting 'adaptive behaviour' by providing dynamic techniques and tools are: General User Modelling Shell (GUMS) [15], UM User Modelling Toolkit [40], User Modelling Tool (UMT) [33], Belief, Goal and Plan Maintenance System (BGP-MS) [41] and Prolog based Tool for User Modelling (PROTUM) [43].

Despite the above-mentioned early research works on user interface adaptation techniques, Akoumianakis and Stephanidis [49] identified limiting capability drawbacks in the behaviour of these adaptive user interfaces in terms of co-operation and intelligence. These drawbacks can be categorised into three spheres of limitations:

(i) Inadequate support in adaptation development of the environment of the user interface;
(ii) Working within a finite and hence bounded and limited modelling space — this restricts severely the ability for adaption beyond a narrow range of environments and situations;
(iii) Pre-determinancy that should be used more intelligently when performing the adaptation.

Whilst it may be easier to build an adaptive system from the beginning, there were no precedents for methodology in modifying non-adaptive systems into an adaptive system at the time when Savidis and Stephanidis [149] conducted their study.

A partial solution can be achieved by the application of *a priori* knowledge in the targeted selection of highly probable "adaptations of lexical attributes of abstract interaction objects". Based on this approach, the USE-IT [58,169] tool was devised by Savidis et al. to offer high-level support for authoring the adaptable user interface.

Adaptive behaviour, whilst gaining in popularity, does place heavy demands on both the hardware and the software design cycle, requiring appropriate interface architectural patterns to be incorporated from the very beginning.

An adaptive user-interface composition has these properties [149] :

(i) The profile being executed will change the user-interface during runtime;
(ii) Some components may be discarded as the user-interface dynamically adapts with the needs of the user, which may mean better alternatives being employed.

The transformation of a static user-interface into an adaptive one for legacy code was implemented [149] by taking a stepwise approach of upgrading the relevant class structures. This necessitated that a collection of alternative components needed to be built up from which to select the runtime component selection.

A user-interface refactoring [95] process was used to allow adaptive behaviour with existing non-adaptive user-interface systems. Refactoring is a software engineering process describing the application of source-level changes at the micro-scale level to improve a system's software without modifying the functionality at the domain-specific level [96].

The benefits of their work, as stated in Savidis and Stephanidis [149], are as follows:

(i) The introduction of a process to turn non-adaptive interactive systems into those which exhibit adaptive behaviour; and
(ii) Achieving an adaptive system by refactoring whilst keeping the original structure unaltered.

How a user interacts with an interface is dependent on both the skill and experience of the user. This fact can be exploited to offer the user either a "novice" UI or an "expert" UI to increase the efficiency of the user's performance by means of the AUI.

It is envisaged that an AUI which dynamically changes from novice to expert mode could possibly improve the users' performance.

## 6. Achieving universality through adaptive user interfaces, from different perspectives

This section reviews the earlier research and projects working towards achieving universal usability, from different perspectives, through adaptive user interfaces. The principal aim was to evaluate them to identify a direction to achieve cross-cultural usability through plasticity of user interface design. The review has followed the evolution of this research. The look-up table (Table 3) demonstrates the aspects/sections that are covered in the review:

### *6.1. Universal usability for multiple Contact Centre Agents (CCA)*

The main point of contact between customers and companies is through the company contact centre. It is thus vitally important to keep the customer satisfied by offering a timely service; this underscores the importance of fast information retrieval by the contact centre agents (CCAs) when customers need answers to their questions. But the CCAs themselves have varying skills and experience and this is not helped by the high staff turnover in this sector. Empirical evidence exists that shows performance increases when the computer UI best matches the user's level. This clearly means the software system has to offer customised

**Table 3**
Look-up Table.

| § | Title |
|---|---|
| 6.1 | Universal Usability for Multiple Contact Centre Agents (CCA) |
| 6.2 | Runtime Model-Based Approach in Designing AUI |
| 6.3 | AUI for Large Scale Software Projects |
| 6.4 | User Modelling for Industrial Control Applications |
| 6.5 | Adaptive Personalisation for Services Offered by Mobile and Wireless Devices |
| 6.6 | AUID for Airborne Crew Stations |
| 6.7 | Adaptive Web Site |
| 6.8 | Adaptive Hypermedia |
| 6.9 | User Characteristics and User Customisation |
| 6.10 | Design and Development Tool to Aid Adaptation |
| 6.11 | AUI for Mental Workload Dynamics |
| 6.12 | AUI for Generation of Web Services and Construction of Query Forms |
| 6.13 | Universal Usability for Adaptive Vehicular Communication System |
| 6.14 | AUID for Multimedia Courseware |
| 6.15 | Use of Fuzzy Approach |
| 6.16 | Adaptive Interface for Conversational Speech Processing |
| 6.17 | AUI for Multi-modal Interfaces |
| 6.18 | Performance Evaluation of Plasticity of User Interface |
| 6.19 | AUID for Spatial Applications |

and individualised interfaces to increase performance efficiency and hence productivity of the user.

A significant study [145] concentrated on the application of AUIs in the work of CCAs and whether the application of AUIs in fact improves interaction with the UI. The primary objective of this research [145] was the introduction of an AUI model in the field of contact centres (CC). A proof of concept was thus created and evaluated: the results established the success of the adaptation through the subjective metric of greater satisfaction and the improved performance of the CCAs.

The requirement for user training will also be reduced when a good UI design is adopted that accords with user-centred design principles. This strategy should hence be adopted in designing the AUI in CC domains.

No currently existing software was found that supported nor catered to the different expertise levels of the CCAs. The use of AUIs is one way to address this deficiency. The basis of an AUI is that it can dynamically change, based on certain characteristics of the user. Consequently, an improved customer experience is provided by using a more customised UI directly related to the skill level of the user by the application of an AUI. The objective of this research was to see whether an Intelligent User Interface (IUI) model for CCAs could be combined with existing AUIs and, furthermore, to see if this combination would also contribute to an increased productivity and usability of CCAs. For CC computer UIs, Jason et al. [145] developed an AUI model.

### 6.2. Runtime model-based approach in designing AUI

In order to develop multimodal and context-sensitive interfaces for applications, which are adaptive, a runtime model-based approach has been used by Blumendorf et al. [142] to formalise the design, map the state of the interactive system and to automate the modification of the UI configuration. The entire process thus requires a comprehensive knowledge of the interaction state and the current context. This can be more elegantly achieved by constructing models of the users, environments and platform [82], including the UI's current state and design.

The traditional approach in designing AUIs needs all the possible user interfaces which are likely to be used to be defined during the design phase, along with the user environment. Two such examples of early projects are: USer Interface eXtensible Markup Language (UsiXML) [104] and TERESA (Transformation Environment for inteRactivE Systems representAtions) [97]. It should be noted, however, that UsiXML and TERESA focus especially on modelling languages. Blumendorf et al. [142] point out the restrictive nature of this approach, as this makes the UI design needs inflexible. Subsequently the suggestion was made that a future AUI system should be designed such that the UI is both flexible and adaptable to runtime parameters (i.e. adaptive). This approach was taken as an optimisation problem in the SUPPLE project [94,144], which adapted widgets and their arrangement. SUPPLE is specifically for end users with motor impairment and belongs to the class of ability-based adaptive user interface.

Recent examples of "dynamically runtime adaptable" (i.e. adaptive) AUIs include the works of: Stephanidis et al. [54], Balme et al. [90], Clerckx et al. [107], Clerckx et al. [122], Duarte and Carriço [108]. In the class of dynamically runtime adaptable approaches, the design decisions are not hidden in the generated code. Sottet et al. [5] further enhanced this approach by suggesting that the models be maintained to stay "alive" at runtime. This allowed the design rationale to be more easily and immediately available and accessible for the real-time plasticity of the user interfaces. Multimodal Adaptive Interaction for Smart Environments was also proposed by Blumendorf et al. (November 17, [131]).

### 6.3. AUI for large scale software projects

Large scale software projects such as enterprise applications, for example ERP (enterprise resource planning) and CRM (customer relationship management) may consist of millions of lines of code. Despite their sophistication and scale, they are not generally designed with an AUI interface, even though they may have thousands of user interfaces to be used by a wide spectrum of users with varying skills, cultures and similar distinguishing attributes. Akiki et al. [161] were the pioneers in adopting an AUI approach for such enterprise software. Their specific contribution was in "simplifying enterprise application UIs through engineering adaptive behaviour". This definition can be provided practically through the use of a minimal feature set and optimised context-of-use based layout, determined, for example, by the culture [156] and skill [120] of the users.

Akiki et al. [161] developed a "Role-Based UI Simplification (RBUIS)" tool based on their CEDAR architecture [159,160], as shown in Fig. 4, which shows the steps involved in the simplification of the UI enterprise application by adaptive behaviour based on engineering roles. The CEDAR architecture, being dynamic at runtime, offers greater flexibility when performing the actual UI adaptations. This makes it particularly suitable for its application in offering a generic service-oriented solution using various technologies of APIs, including RBUIS.

Of particular concern is the need for support tools to enable adaptive interfaces for developers and I.T. professionals. The CEDAR reference architecture (Fig. 3) is an IDE (Integrated Development Environment), which offers this provision for "visual design and code editing tools for UI models" [159,160]. The CEDAR Studio interface was however identified as being in need of major modification, with the need to implement code views to support larger AUI models using (i) XML; (ii) UsiXml [104].

The first probable use of an artificial neural network (ANN) to design a multimodal user interface system (MUIS) prototyping tool was described as early as 1995 in Arcand and Ramstein [38]. The major limitations included:

(i) Finding that the lack of experience of the user negatively affected both the comprehension of the information provided by the force-feedback interface and the errors committed whilst using it.
(ii) Situations that caused instability in the interface, being a feedback system, which greatly increased the error rate and fatigue of the user.

### 6.4. User modelling for industrial control applications

User modelling for industrial control applications was described by Averbukh et al. (12–15 October [50]), with emphasis on taking into account individual user characteristics. This was finely tuned to encompass such factors as the information processing due to the user's brain hemisphere asymmetry. In particular, Averbukh et al. (12–15 October [50]) also identified the need for further research on "domain-independent user-, interface- and interpretation models and game-controlled interactive facilities".

### 6.5. Adaptive personalisation for services offered by mobile and wireless devices

Billsus et al. [77] focused on mobile and wireless devices and correctly identified the need for adaptive personalisation for services offered through these types of devices. Even though no smart phones were used at that time, some phones were beginning to have screens for extended amounts of text and some modest graphics: the needs for these are basically the same as for smart devices. Formatting of the data in terms of content and resolution is needed for mobile devices for easier interactivity, especially taking account the small screen and restricted human input modality. The study found:

(i) Wireless providers in general offer little personalisation of their content for users.
(ii) Only technically proficient users customise the content that they receive.
(iii) The customisation options offered are usually very limited.
(iv) Regular maintenance of the user profiles is necessary — this being only practiced by the adept users.
(v) The user experience differs from that when accessing it via wired means.

To accommodate these varied behavioural patterns of the users, the researchers strongly advocated the adoption of an "automated approach" in delivering a personalised UI based on AI (Artificial Intelligence) and statistical techniques. They further stated that this automatic customisation should be based on the actual user behaviour and not just their feedback.

### 6.6. AUID for airborne crew stations

Haptic input devices were considered by Brickman et al. [46], who focused on developing an adaptive interface with force-reflective, haptic stimulation that could be used to provide individuals with information about their location and movement through space. The research also considered the adaptation to the changing psychological and physiological states of the user, which are both highly influenced by culture.

The work of Brickman et al. [46] also found that, for implementing a realistic flight simulator control and environment, it was essential to employ haptics, especially a force feedback control stick that feeds back the effects of turbulence. However, the airborne crew stations only found this useful for certain cases. Consequently, refinements of the use of haptics to model other flight scenarios or aviation tasks were also identified. For simplicity and generality, they had to adopt a sub-optimal approach, however.

### 6.7. Adaptive web site

In 2000, Perkowitz and Etzioni [62] proposed four factors that needed to be considered in the design of adaptive web sites, these being the:

(i) Degree of automation;
(ii) Type of adaptation, such as links, pages, format of the page and the information;
(iii) Access or content-based adaptation and
(iv) Transformation or customisation of information.

These factors have all been subsequently adopted in AUI systems.

The use of colour and layout together in UA design were found to maximally shorten the time to complete tasks interactively [78] rather than just using each adaptation individually.

### 6.8. Adaptive hypermedia

Brusilovsky [70] give a clear definition of "adaptive hypermedia systems" as the discernment and then construction of "a model of the goals, preferences and knowledge of each individual user, and [the] use [of] this model throughout the interaction with the user, in order to adapt to the needs of that user". The move away from a static hypermedia interface where the pages served by the web server are static to one that dynamically changes to cater for the needs of the user is essential for the interface to be truly "all things to all people".

Six types of adaptive hypermedia were identified in 1996 by Peter Brusilovsky [47,70], a leading researcher of this field. These six types are:

(i) educational hypermedia;
(ii) on-line information systems;
(iii) on-line help systems;
(iv) information retrieval hypermedia;
(v) institutional hypermedia
(vi) systems for managing personalised views in information spaces.

The adaptive hypermedia system must consider the user's knowledge and goals and offer appropriate, relevant and useful hyperlinks wherever and whenever culturally necessary [47]. This means that it must occur at the most pertinent point in time of the interaction and must consider the innate cultural assumptions of the end user.

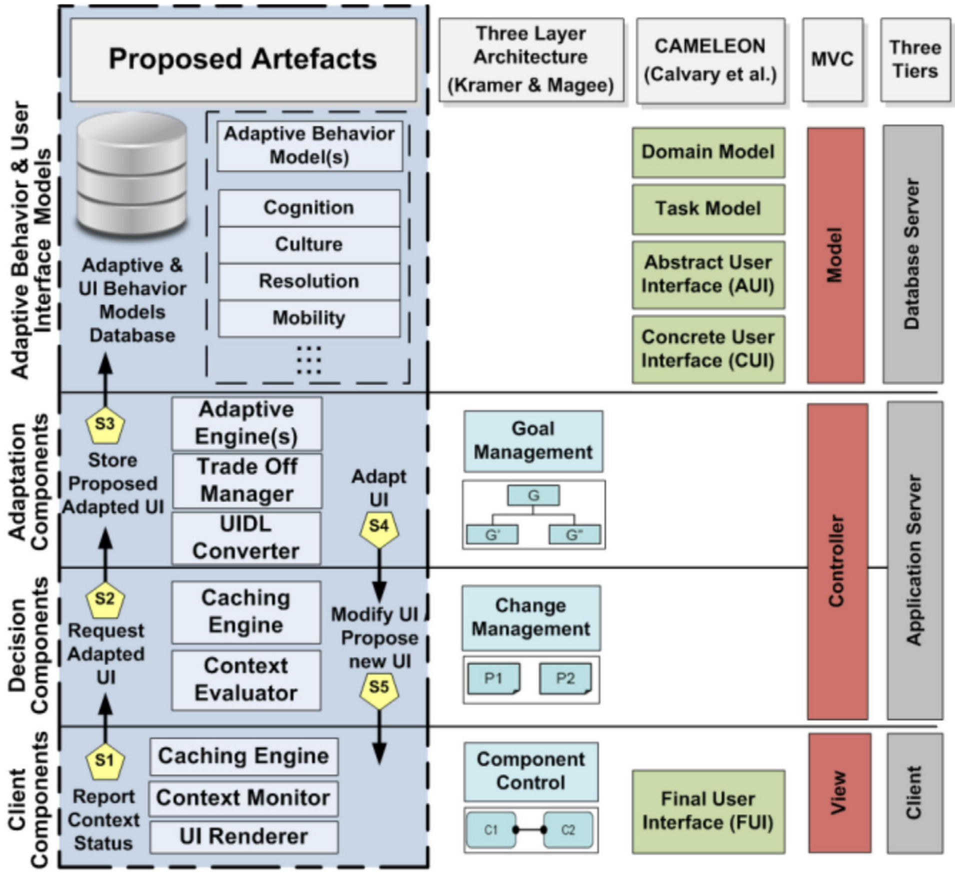



**Fig. 3.** The CEDAR Architecture [159,160].

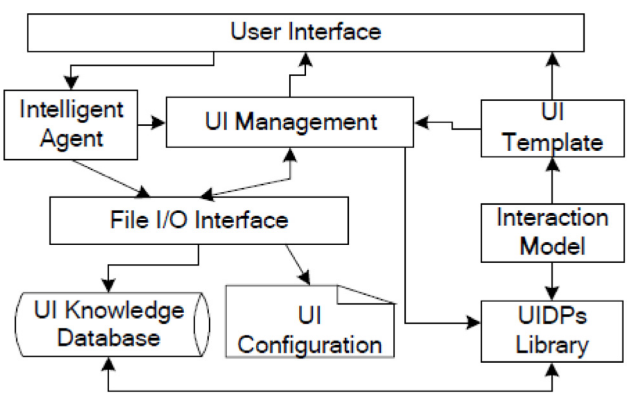



**Fig. 4.** The agilely adaptive user interface model based on UIDPs [109].

### 6.9. *User characteristics and user customisation*

An interesting article by Feeney and Hood [11] points out the work of an early visionary, Licklider [8], as far back as in 1960, where he proposed that computers should take into account the ability of the user on a task-by-task basis. Further, he suggested distributing the tasks between the user and the computer, based on their respective abilities.

Mason and Mitroff [10] took another approach by suggesting "a structured context for the design of a Management Information System" (MIS) based on how the user ("the manager") performed "on a test". Lucas [9] took a different approach by focusing on the actual needs of the user through the utilisation of a detailed questionnaire.

The underlying foundation in the studies mentioned above, was to account for the individuality of each user and thus make the UI truly adaptive. An important point raised by the researchers was to question how the user actually "deals with the information presented [to them] by a computer".

In Feeney and Hood [11], a correlation analysis was carried out to find out the relation between the 'Format Instrument' (computer-like displays) and seven 'Psychological Style Instruments', which are: Personality Traits, Vocational Interests, Cognitive Style, Perceptual Style, Vocational Aptitude, Computer Experience and Application Experience.

Feng et al. [109] offer a model to make the AUI truly agile, as shown in Fig. 4. The key components of the model's various UI functional blocks are clearly shown, including the use of the Intelligent Agent. By configuring and extracting the information from the UI Design Patterns (UIDP), the UI can be generated more quickly. The user driven changes for the UI directly alter the parameters of the corresponding UIDP, which then feeds back these changes to the UI through its Knowledge Database (UIKD).

To further enhance user customisation, Bunt et al. [91] proposed the use of cognitive modelling to take into account the behaviour of the user but noted that the implementation efficiency could vary. A specific language for adaptive decision making was suggested in Savidis et al. [105]. This language was able to process the user behaviour during runtime and dynamically customise the UIs. The quick acquisition of an adaptive system by these methods was, however, concluded to be unlikely [109].

Adaptive personalisation, though offering substantial performance benefits for the user, had one major drawback. This was that it was initially devised largely for large-screen non-mobile office or desktop machines. Using "high accuracy adaptive menus"

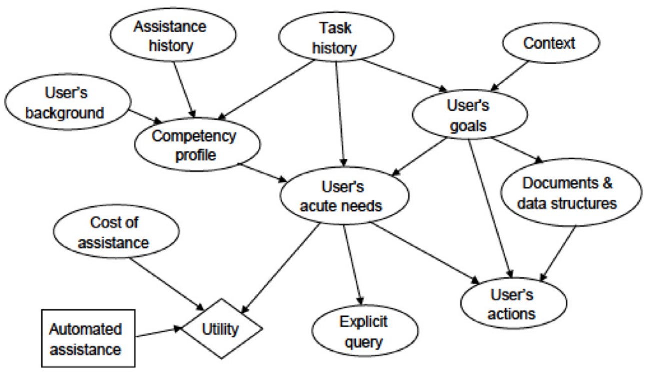



**Fig. 5.** An influence diagram for providing intelligent assistance given uncertainty about a user's background, goals and competency in working with a software application [52].

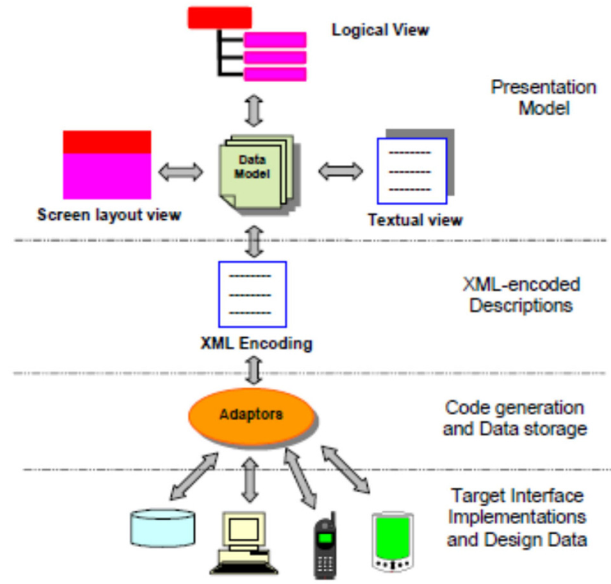



**Fig. 6.** Adaptive GUI design tool architecture [86].

for smaller screen devices further increases the performance benefits when utilised on smaller screen devices. This too has a drawback; however, as the user may miss out on the many other features of the system (Findlater and McGrenere, April 5–10, [123]). The experimental research by Findlater and McGrenere differentiates between the menu structure displayed for the small screen and that for the large screen. The menu is broken into an adaptive top section and scroll widgets for the small screen. For the large screen display, the menu is solely static.

McGrenere et al. [118] elucidate two ways of aiding personalisation by having: "system-controlled adaptive menus" as well as "user-controlled adaptable menus". Their tested prototype showed the superiority in terms of navigation and learnability of having user-controlled adaptable menus over system-generated adaptive ones. They point out that other features of the interface beyond just the menus and toolbars also need to be considered.

Mejia et al. (03–06 September, [158]) propose an adaptive user interface that attempts to create an interface on a human model that is as complete as possible, based on such parameters as the psychological, physical and cognitive human characteristics, plus consideration of the user's demographic and experience. Moreover, this adaptive system can be integrated with adaptable elements via user feedback to make the interface more personalised.

An early literature review [17], based on about 50 research papers, concluded that an adaptive interface must include a knowledge base consisting of four domains: current user, interaction scheme, problem task and underlying system. Granić, A. and Nakić (25–28 June, [116]) give the empirical study methodology, taking into account the user characteristics, the knowledge gained through an e-learning system and how these factors affect interaction.

The Lumière Project [52] utilises probability and Bayesian techniques to provide computer software assistance based on the user's background, queries and actions. Uncertainty in the complete knowledge of the background of the user is considered by using "influence" and "intelligence assistance" factors in the algorithm, as shown in Fig. 5.

A study by Miraz et al. [162,163] concluded that more consideration should be given in creating the text, especially taking into account the cultural-linguistic heritage of the users and the application of graphics in websites which are multilingual.

### 6.10. Design and development tool to aid adaptation

The use of a Visual Language Generator (VLG), to implement grammatical inference techniques, for designing AUIs, is described in Crimi et al. [19]. The VLG, though very powerful in terms of helping the programmer to customise the AUI, can be further extended to be made even more applicable to any visual environment by the use of the semantics-based inference methodology [19]. Ferrucci et al. [48] further generalised this by allowing the end-user to select the most appropriate algorithm for the particular application environment, based on the 'General-Inference' (Gen-Inf) scheme. This scheme was devised by these researchers by formulating commonalities between popular syntactic inference algorithms. The algorithm was further enhanced by the adoption of a 'semantics-based inference methodology' that complied with the scheme. The algorithms studied by the researchers were modified to exploit "the semantic similarities between symbols" stipulated by the designer for the application. This helped to attain a more generic visual language model where the user could select the Gen-Inf compliant algorithm to maximally match the user requirements in the application.

With the proliferation of different types of computing devices, support for multi-device adaptive interfaces is ever more necessary. Grundy and Yang [86] describe the creation of such a multi-device adaptive interface using a Java Server Page implementation, called the 'Adaptive User Interface Technology' (AUIT). Fig. 6 shows the Adaptive GUI design tool architecture used to carry out this task, based on the XML language. Fig. 7 shows the tiled window-based approach used that follows closely the AUIT XML screen-based structure [86].

### 6.11. AUI for mental workload dynamics

In 1988, Hancock and Chignell [13] proposed an Adaptive User Interface for Mental Workload Dynamics. The concept of mental workload [13] and stress must be taken into account to reduce user error and increase the accuracy and, more importantly, to avoid potential catastrophic incidents occurring through human error, especially in safety critical applications. Designing AUI to take these factors into consideration can help control their effects and delegate tasks more equitably between the different users. The mental workload concept can be visualised as a three-dimensional graph, with the axes representing the: effective time for action, perceived distance from achieving the goal, and the

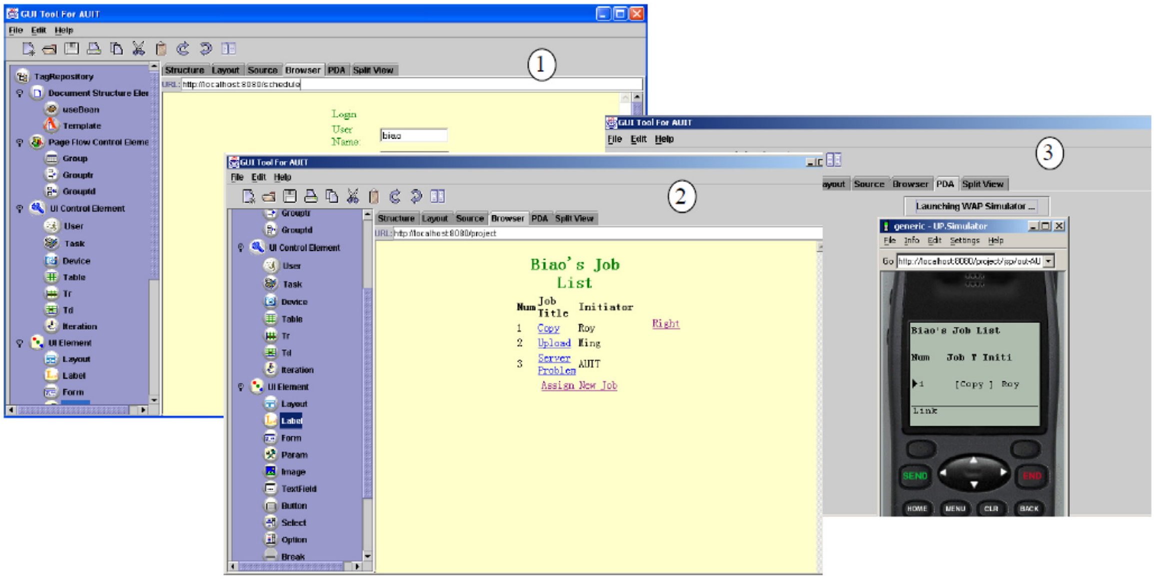



**Fig. 7.** Examples of running interfaces generated by AUIT [86].

level of effort required. The distance from the origin represents the level of mental workload.

Studying how the human mind works and applying this to design HCI interfaces is valuable to reduce, for example, human errors due to tiredness or inattentiveness [76]. This is the premise of the application of cognitive models in HCI design.

Consequently the cognitive models are applied to the HCI interfaces to enhance human performance and also to constrain it where it is deemed beneficial to performance [63]. The model is used to enhance performance over time and to predict when lethargy is setting in, thus potentially preventing fatalities in safety-critical applications from occurring.

It is crucial to understand that cognitive architecture (Byrne, 22–27 July 2005) is often disparate, concentrating on modelling one human attribute such as memory or vision. However, HCI needs to take a holistic approach and thus the cognitive architecture needs to integrate a more complete model of the human sensory system and then apply this to enhance HCI interfaces.

### 6.12. AUI for generation of web services and construction of query forms

Papers by He et al. (24–26 October, [117]; 23–26 September, [125]) propose a framework, as shown in Fig. 8, for web services to be generated by AUI. This takes the Web Service Definition Language (WSDL) to output the customised Graphical User Interface (GUI) for the specified device and platform, thus saving considerable coding efforts and helping to generate better GUIs.

Perugini and Ramakrishnan [113] identified the issues relating to "Interacting with Web Hierarchies" by comparing the different related methods and proposed solutions, including an adaptive interface design. They concluded that the key important features, that need to be incorporated in information hierarchy interfaces, are:

(i) Navigational order flexibility;
(ii) Dependencies, exploration and exposition;
(iii) Procedural tasks support and
(iv) Provision of features used.

Jayapandian and Jagadish [136] proposed automating the design and construction of query forms using an adaptive algorithm. The importance of this work is based on two factors:

(i) query forms are recognised as the most user-friendly way to implement the task of querying;
(ii) query interfaces are of paramount importance in assessing the usefulness of a database.

Hence combining both features into a forms-based interface takes advantage of both factors. However, the design of a form is quite an arduous task. A tunable clustering algorithm was used by Jayapandian and Jagadish [136] to implement an automatic form-generation interface, applicable to a wide range of queries and databases. Even though requiring minimal user input, human assistance can still lead to further accuracy.

### 6.13. Universal usability for adaptive vehicular communication system

A Conceptual Framework for Agent Interaction in Intelligent Adaptive Interface (IAI) Design was proposed by Hou et al. [154]. IAIs are being applied to "complex sociotechnical environments" such as for control of Unmanned Aerial Vehicles (UAVs). A standardised approach is required to minimise the cost and increase the effectiveness in the interface design. A generic user-centred approach for IAIs, as shown in Fig. 9, is espoused that makes use of Adaptive Intelligent Agents (AIAs) with the aim of maximal performance of the system. The focus on the operator can clearly be seen in Fig. 9.

Tchankue et al. (30 November–2 December, [157]) describe an adaptive vehicular communication system, called MIMI (Multimodal Interface for Mobile Info-communication), the architecture of which is shown in Fig. 10. The MIMI system used neural networks and such inputs as the steering wheel angle and velocity of the vehicle to maximise the safety of the driver. This was achieved by reducing the "cognitive workload" by preventing activation of services which could have distracted the driver's attention, such as suppressing phone calls whilst driving. Further understanding of the driver behaviour can be enhanced using fuzzy logic tied with neural network processing to avert or warn of impending critical situations.

Rudin-Brown [148] studied the techniques of limiting behavioural adaptation (BA) for in-vehicle intelligent transport systems (ITS) using feedback through adaptive design, as shown in Fig. 11. The primary purpose of this system is to protect the driver and ensure safety. Whilst adhering to cultural sensitivities, this

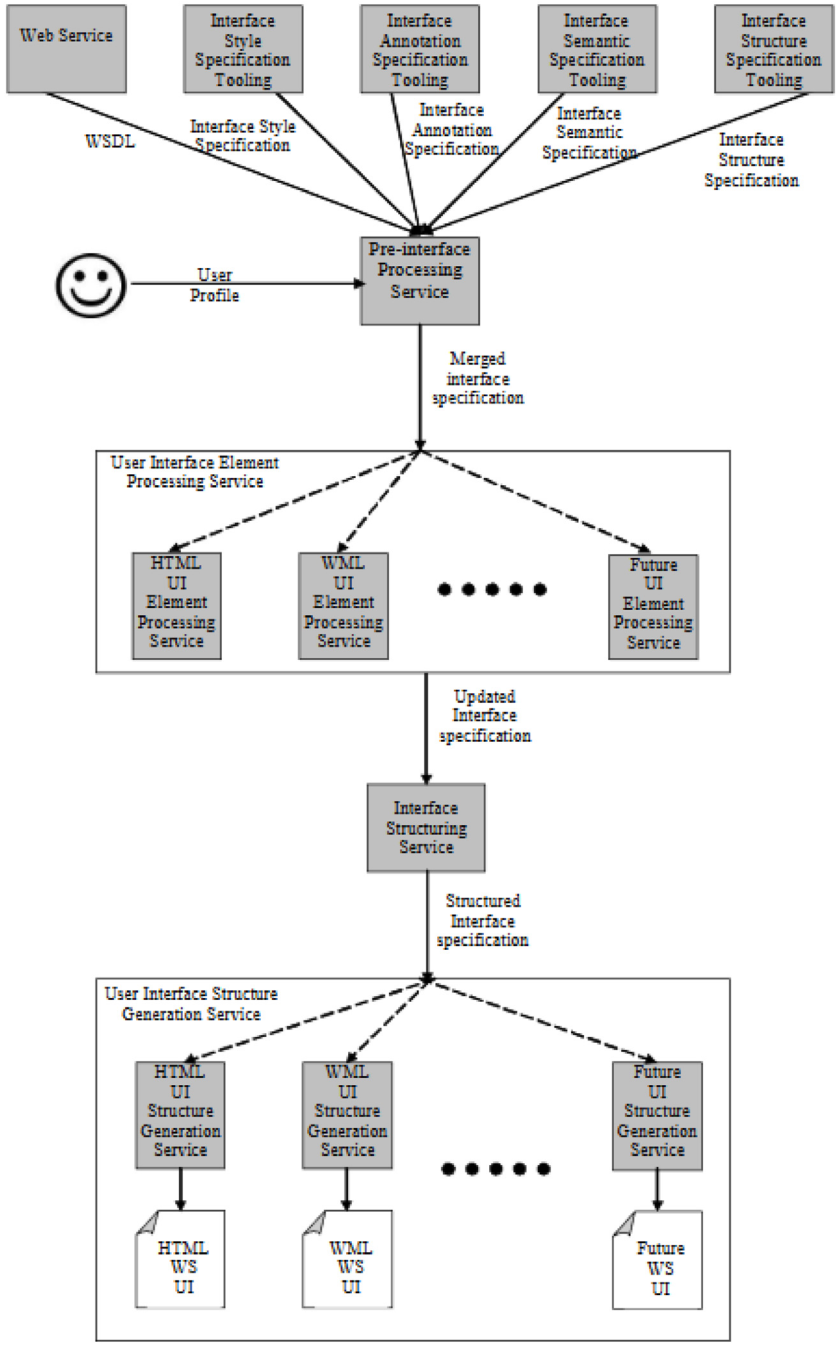



**Fig. 8.** Framework architecture overview (He and Yen, 24–26 October, [117]).

can be achieved by applying the principles of both "intercultural adaptability" as well as "adaptive interface design" at the design phase level of the development process.

The need to develop a well-defined model of human–computer interaction (HCI) in a vehicular environment was stated by Moray (August [56]) as the key foundation for:

(i) better communicability between the various engineers and in understanding the psychological factors contributing to the behaviour of the driver;
(ii) predicting the likely future behaviour of drivers and
(iii) designing a better automated ITS system.

Lavie and Meyer [147] discussed both the positive and negative factors of using AUIs in a vehicular environment, based on levels of adaptivity, as shown in Fig. 12. Four factors were identified as a function of their vehicular telematics system, these being the:

(i) four levels of adaptivity, ranging from manual to fully adaptive;
(ii) range of tasks;
(iii) familiarity of the routine and
(iv) age of the user.

They concluded that adaptivity was not always useful and that it depended on various human factors such as age and the difficulty of the task — which involved a highly increased cognitive workload. Adaptivity was, however, useful where the routine was identified as being familiar. Intermediate levels of

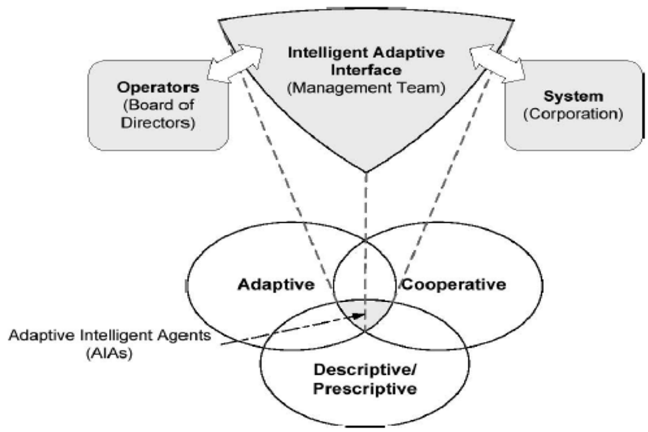



**Fig. 9.** Operator–agent interaction model [154].

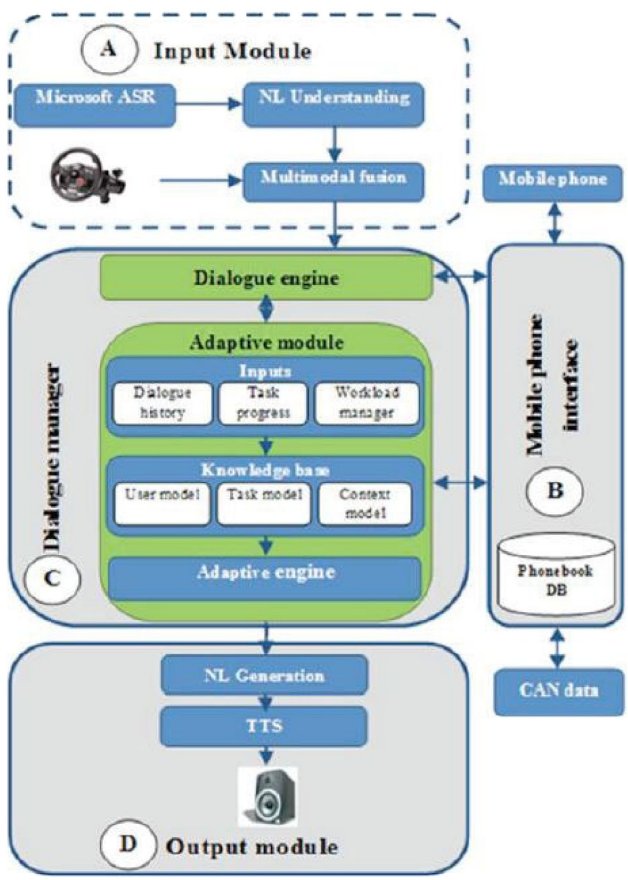



**Fig. 10.** Architecture of MIMI (Tchankue et al., 30 November–2 December, [157]).

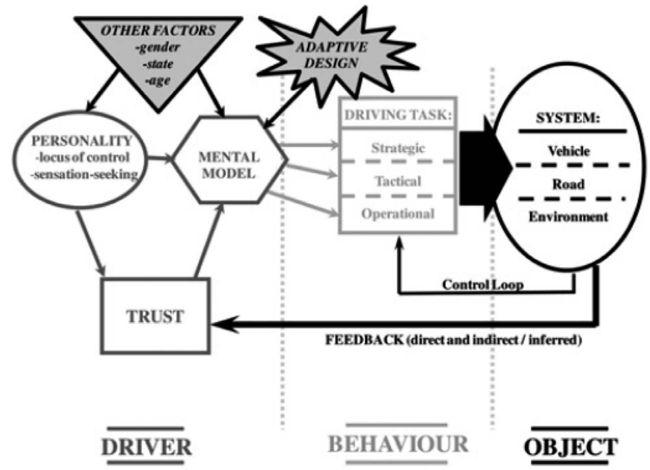



**Fig. 11.** Qualitative model of Behavioural Adaptation (BA) [148].

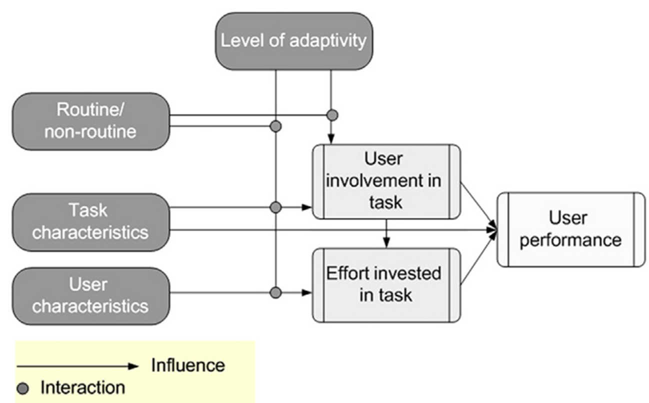



**Fig. 12.** A schematic representation of the levels of adaptivity [147].

adaptivity were found to be good in general for enhancing the learning of the user.

### 6.14. AUID for multimedia courseware

Muda and Mohamed [110] introduced the application of adaptive and adaptable techniques while developing multimedia courseware. The prototype developed was used for primary school children in Malaysia to teach religious modules. As shown in Fig. 13, their implementation of the educational adaptive courseware integrated both a graphical and an interaction-based approach.

### 6.15. Use of Fuzzy approach

The main premise of "fuzzy logic" concerns the handling of imprecise or ambiguous data. An early adoption of fuzzy logic to implement an adaptive user interface was proposed in 1994 by McNeill and Thro [35]. Their fuzzy logic adaptive interface engine controller is shown in Fig. 14. This concept was applied to process the user's shopping preferences (Nyongesa et al., 13–17 October, [87]) with the use of information filtering to further enhance the accuracy: the subsequent product information was then presented to the user.

Nyongesa et al. (13–17 October, [87]) presented the use of fuzzy logic to enable adaptability in the web interface for online shopping, based on the user's shopping history. It was concluded that using fuzzy logic was in general a viable proposition. Fig. 15 shows a generalised model of the adaptive system adopted.

### 6.16. Adaptive interface for conversational speech processing

Oviatt et al. [99] of Oregon Health & Science University is one of the pioneers in introducing an adaptive interface that processes conversational speech. Designing a highly accurate machine-based conversational speech system is not a trivial task due to

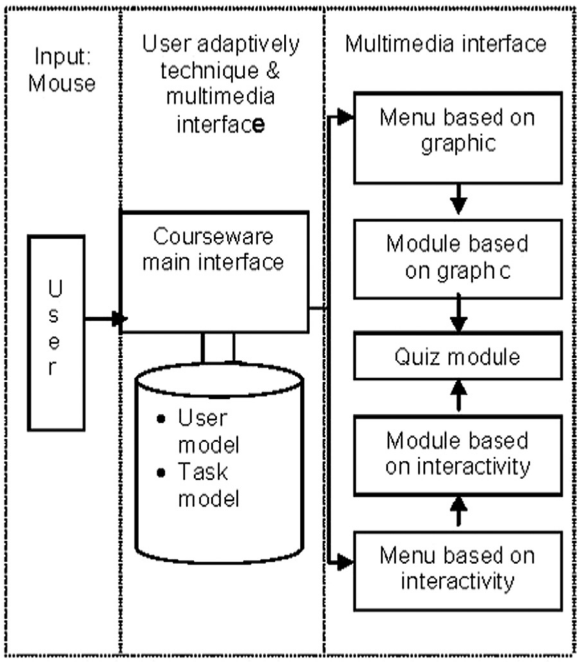



**Fig. 13.** Courseware framework [110].

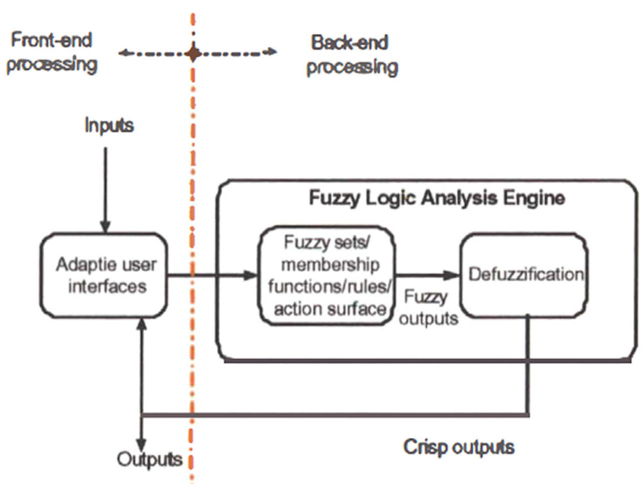



**Fig. 14.** Fuzzy adaptive interface controller [35].

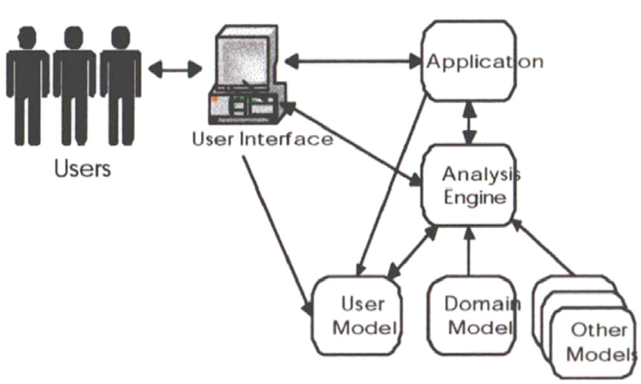



**Fig. 15.** An archetypal system employing a user model (Nyongesa et al., 13–17 October, [87]).

the almost infinite variation in human speech parameters. Using a machine text-to-speech (TTS) system to train human speakers to spontaneously adapt their speech to the system speech greatly increased the accuracy of machine voice intelligibility. Accuracy was further enhanced by guiding the human speakers to interact within the system speech processing margins. They further identified the need for improved and accurate "predictive models of human–computer communication" for more accurate conversational machine interfaces using a more adaptive and user-centred approach. Contemporaneously, multi-modal conversational interfaces with non-voice input modes using touch-enabled devices or pen input devices were already appearing with greater functionality and expanded flexibility in comparison with voice-driven-only language interfaces [98].

### 6.17. AUI for multi-modal interfaces

Prammanee et al. [111] present an "adaptive multimodal interface design technique" for diverse mobile devices within the constraint of a limited screen display size. Furthermore, services offered were tailored to the hardware and software capabilities of the device. Their scheme concentrated on both the requirements of the application as well as of the user needs. The multimodality present in a typical mobile environment addressed is shown in Fig. 16. This architecture is called MID-B ("Multi Modal Interface-Binding Engine"). Calvary et al. ([82]; Thevenin et al., 18 August [106]) deal with plasticity for the development of plastic user interfaces especially in the domain of multimodal interaction.

Oviatt [88] presents a review of multimodal interfaces and suggests extension of this to adaptive multimodal–multisensor interfaces. This, however, requires the construction of a complex human model that not only processes the voluntary gestures and movements but also the involuntary reflexes as well. This also requires taking a multi-sensor fusion approach in order to predict the likely future behavioural pattern of the user, often in near real-time. Complexity may also be reduced by selectively choosing the most appropriate methods and measures.

### 6.18. Performance evaluation of plasticity of user interface

An analysis (Paymans et al., 13–16 January, [100]) of context-aware user interfaces showed that adaptation mechanisms surprisingly have a cost–benefit trade-off for usability. Providing too much user support though facilitated ease of use, however, had the negative consequence of degrading user learnability, although the sample size only consisted of 17 participants in their "mental model of adaptive system" prototype evaluation process.

Though there may be benefits and disadvantages of adaptive user interfaces, the results of a study (Gajos et al., April 5–10, [124]) showed the greatest satisfaction and performance improvement when increasing the accuracy over the predictability of the AUIs. The task presentation menu is shown in Fig. 17. Furthermore, the great gains in user satisfaction over the reduction in predictability suggest that machine learning algorithms should be considered seriously in the implementation of AUIs.

Reinecke and Bernstein [156] present a culturally adaptive system known as MOCCA (2009) as shown in Fig. 18, which helps users in online task organisation by using a to-do-list. For its time, this was a very innovative and thoughtful step in the evolution of AUI design as the cultural aspect of the user had been neglected previously. Their research found faster (22%) and more accurate interactivity with fewer clicks when using the culturally aligned prototype. The prototype was aligned based on these cultural attributes: country or previous country of domicile.

An adaptive user interface has been developed to enhance the interactive three-dimensional navigation adaptation in virtual

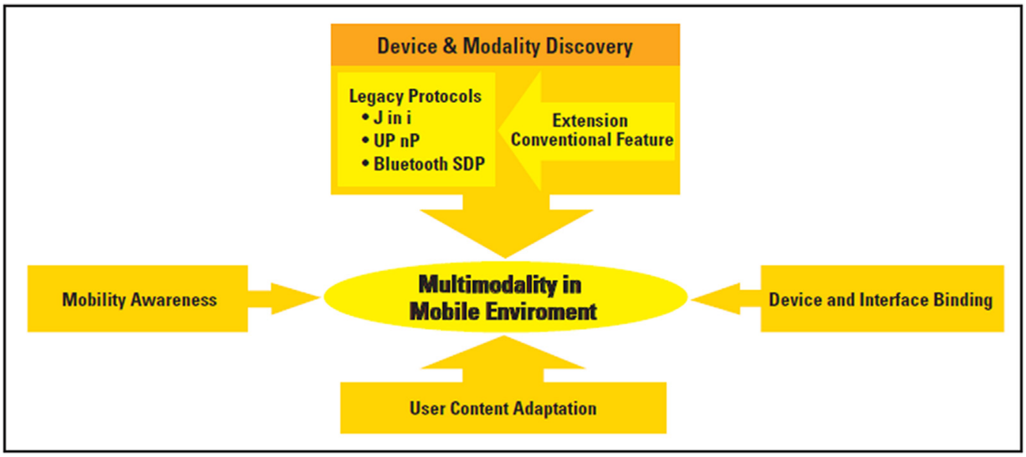



**Fig. 16.** Multimodality in mobile environment [111].

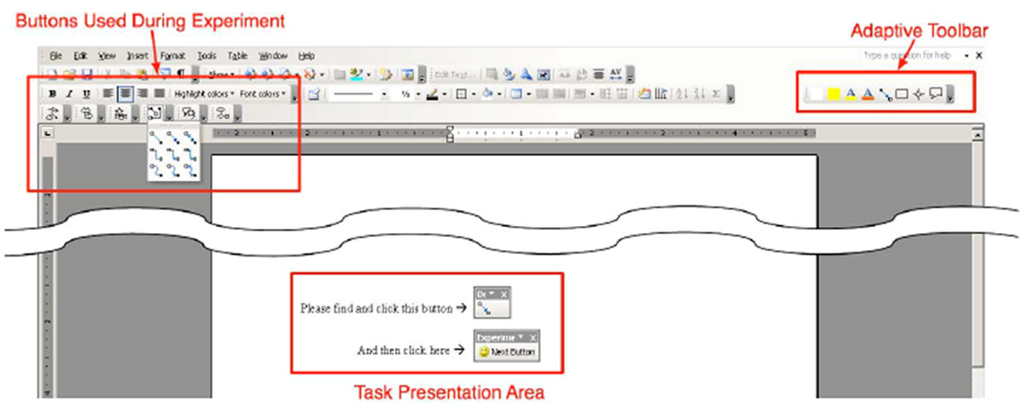



**Fig. 17.** Task presentation (Gajos et al., April 5–10, [124]).

environments (Shi-wei and Shou-Qian, 26–29 November, [139]) using Rough Sets Theory, introduced by Pawlak [12]. The architecture for this is shown in Fig. 19. The success of the system lies in having a realistic user model. They concluded that their system did improve the user performance in 3D navigation.

### *6.19. AUID for spatial applications*

A location-aware and context-sensitive intelligent guide system was reported in [112,140] using a "metal detector" for accurate navigation, driven by user preference settings. The system was successfully tested for urban and tourist use. Past routing history and personalisation were used to highlight points-of-interest to the user, as shown in Fig. 20. The "place learning algorithm" used to achieve this involved first finding past visits within a certain distance from a point of interest (the 'sample visits') then if this point is frequented, the mean of the sample visits and the actual point of interest location should be statistically identical.

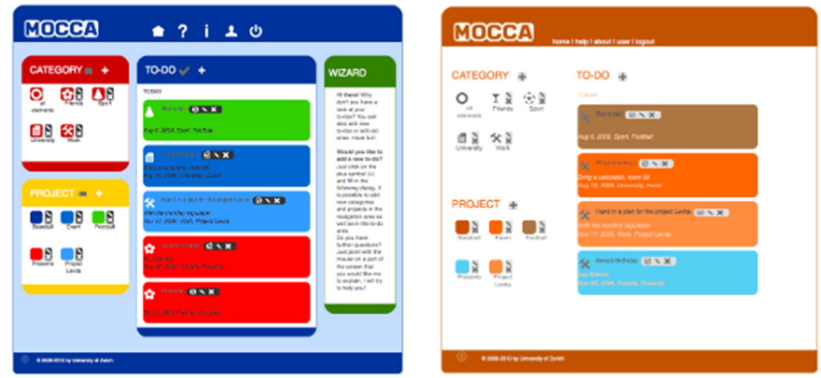
Wakkary and Hatala [121] describe the implementation of adaptive Tangible User Interfaces (TUIs) in the museum environment to appeal especially to young learners. This is based on using a tactile, haptic enabled device with audio, video, text and the control of the ambient environment. The paper concluded that learning is best enabled through "playful interaction". The system architecture is shown in Fig. 21, where location and movement of the user through a museum triggers relevant content.

## 7. Conclusion

This paper presents a literature review of the research related to universal usability, plasticity of user interface design and facilitation of the development of interfaces with universal usability through plasticity of interface design. This survey paper covers the examination of a detailed and comprehensive set of studies of adaptive and adaptable interfaces for universal usability, using 165 research papers spanning over fifty-five years.

The foundations of adaptive user interfaces (AUI), also known as Intelligent User Interfaces (IUIs), have been presented: the raison d'être for IUIs is that they must improve user–machine interactivity. This is presented as a discussion on the motivation and rationale for their use, including the current advances in adaptive interfaces, including their analysis and the challenges they face. AUIs require knowledge from a multi-disciplinary field as they also operate in such a domain. However, for the scope of this research, the following three relevant core domains are focused upon: Artificial Intelligence (AI), User Modelling (UM) and Human–Computer Interaction (HCI). An in-depth approach for each is presented including the latest findings, algorithms and architectures for the adaptive user interfaces.

AUI applications have been given to show the latest examples in the recent field of adaptive user interfaces. These examples have covered both complete systems and prototypes. Their merits, limitations and the challenges facing users have also been discussed. Common effective AUI design principles have been elucidated. For comparison of the various AUIs, a proposed taxonomy



(a) A brightly colored version of MOCCA with a flat button navigation. The participant who received this interface was from Mexico, and had shortly lived in Bulgaria before coming to Switzerland.

(b) A version without the colored borders that define the different areas of the interface. It was personalized for an Indian participant, who had lived in France and the US for several years, which reduced the complexity of the interface predicted by our model for a 'pure' Indian version.

(c) A version with the to-dos in list view, so that to-dos have to be expanded to show more information. The interface was generated for a participant from Poland, who had also lived in Ireland and Germany.

(d) MOCCA with pastel colors, as it was triggered for a participant with Russian, Romanian, and Swiss background. Functions (e.g., delete, add, or edit) are always accessible and add to the information density.

**Fig. 18.** Example interfaces of MOCCA generated for different participants [156].

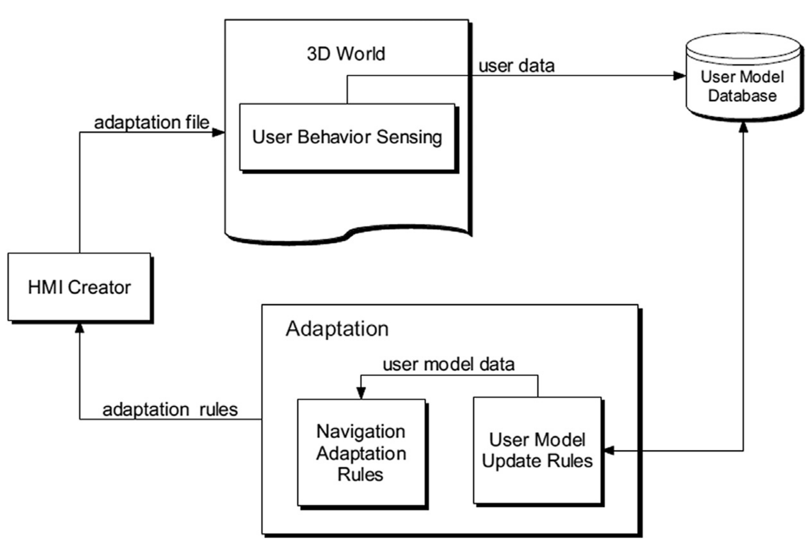



**Fig. 19.** Schema of the architecture for adaptive 3D navigation (Shi-wei and Shou-Qian, 26–29 November, [139]).

was also given. Any gaps identified with the current approaches have also been identified as far as possible. The present research concludes with the challenges still existing, with suggestions for future research to address these.

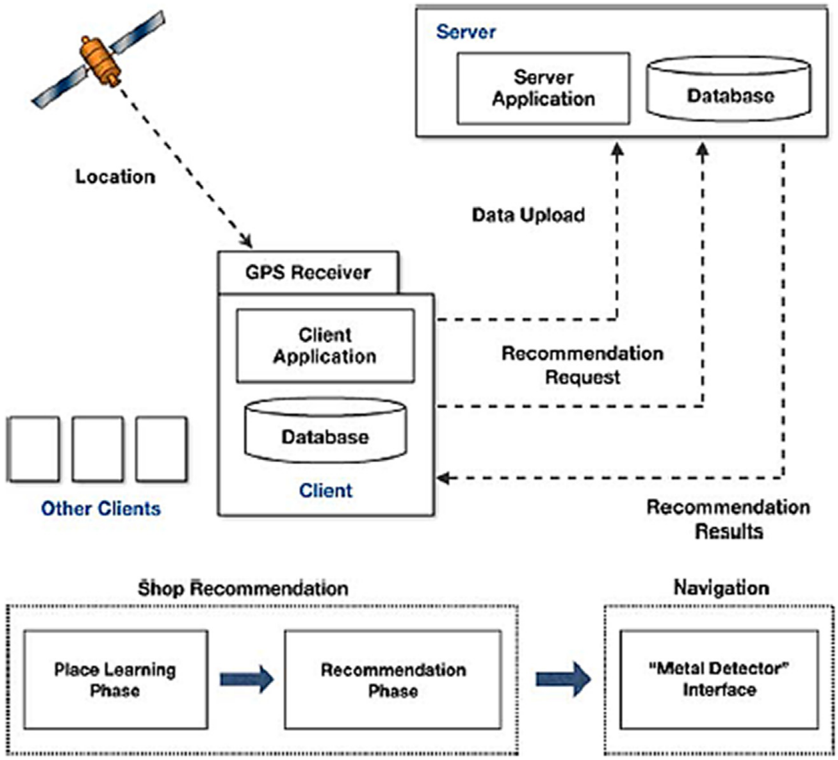



**Fig. 20.** Service procedure [140].

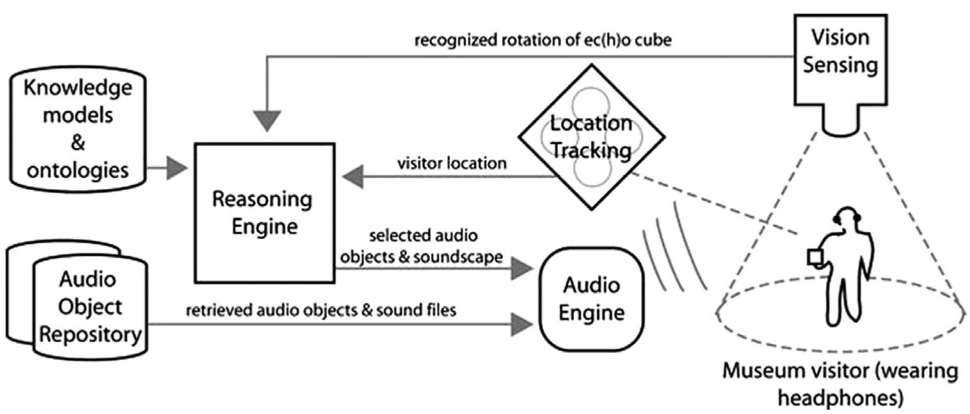



**Fig. 21.** The Tangible User Interface (TUI) architecture [121].

Plasticity of user interface design research is the latest trend in the domain of AUI design, facing many challenges. Some criticism arises from the fact that it is allied with AI, and thus some scepticisms have been raised by the HCI community.

From the literature survey, most of the AUIs are designed for office and web-based applications. Being a machine-learning based algorithm learning from human users, an efficient training vector for fast optimal convergence of the algorithm is a necessity. This can be helped by having a small initial dataset. The key according to the researchers is the bounding of the dataset and not fast computational hardware. The goal is to achieve an accurate user preference model. The peculiar problem with human interaction is that of building such an accurate model from a limited number of datasets obtained from the human user interaction. This problem does not appear to exist in the case of data mining, as vast amounts of data exist, often to be processed offline, for far more accuracy. More research needs to be conducted to ascertain the usefulness and effectiveness of IUIs compared against AUIs: this is due to the lack of a substantial body of research, even empirical research. A notable evaluative empirical research had been conducted in the recent past, but its applicability to adaptive systems should be taken with reservations.

Research in the areas of AUIs has been presented, including their challenges and solutions. It is a complex multidisciplinary subject, but the present work has identified three major approaches. However, the present study shows a lack of interchange of ideas and concepts between the researchers: this is one area that needs to be further encouraged, since IUI research is a multidisciplinary field with applications in a multi-disciplinary environment. New, innovative and disruptive theories and practices need to be applied for a true cross-fertilisation approach.

Licklider, as far back as in 1960, proposed the need for computers to be aware to the needs of the user and to modify their interfaces accordingly. With the mobility of users and mass global market penetration of computing devices, user interface design now has to factor in the abilities of the end users and their different cultures, hitherto scantly regarded. Two factors for consideration are whether the interface should be adaptive under the system control or be made adaptable under the user control: both entail complexities in their implementations.

From the literature survey, the trend is towards an adaptive user interface, but many techniques for this exist. The notable system design components have identified the need for: an "adaptivity afferential component" that stores a priori information about the end user; an "adaptivity inferential component" that determines to what extent the user interface actually needs to be adapted; and finally the "adaptivity efferential component" that actually determines how the adaptivity is applied seamlessly to the system, with the end user's manual input. Another methodology is to use a similar concept to cookies, recording how a user has interacted by using "virtual footprints".

Early pioneering work on adaptive interfaces involved making computer games accessible for handicapped users. However, during this early period, work was hampered by the limited computing resources available at the time. Progress was made initially by the application of a priori knowledge about the user and by refactoring. The driving force behind making user interfaces adaptable was business profit resulting from increasing sales and revenue. If sales agents could more effectively utilise their platforms to sell to a wider range of customers, then this would greatly increase the company's revenue. A runtime model-based approach was seen early on to be best able to offer a context-sensitive dynamically changing interface with the demands of the user interacting with appropriate software routines. Adaptive interfaces were also employed in ERP systems.

With the advent of mobile phones and mobile computing, fast and highly responsive adaptive interfaces were seen as a way to drive the momentum of mobile application uptake, especially in offering personalised services. The provision and customisation of adaptive interfaces was clearly identified as needing to be based on the actual user behaviour and not on their feedback. With the spread of mobile computing devices globally, hitherto formally unrecognised cultural adaptation of interfaces could no longer be ignored. Coupled with a culturally sensitive adaptive interface, a priority was application to e-learning, especially in hypermedia applications.

Current adaptive interfaces utilise the cognitive behaviour of the user, relying on a knowledge base of the user's characteristics. Context sensitive menus may also be offered with localisation and cultural sensitivities considered [170–172]. With increasing computing power came more advanced adaptive interfaces that were more finely matched to the needs of the individual user. This included considering the mental workload and stress level of the user. This is especially important to reduce the error in safety critical applications, such as in driving.

Recent studies have also surprisingly shown that adaptivity is not always desirable. This is the case where age, difficulty of the task and an increased cognitive workload are involved. However, adaptivity should still be applied for routine to intermediate tasks.

Providing the right amount of adaptability in a user interface is a fine balance. Realistic levels of challenge in the learnability for new users should be maintained. The use of machine intelligence to achieve this fine level of customisability was found and "Playful interaction" was found to enhance learnability.

Continuing research is helping to find more and diverse situations where adaptation may be applied. However, its universal application still lies in the future and a universal applicability model continues to be developed. One of the findings is that the AUI properties cannot be ascertained while disregarding the other parameters of the system in which it will be used. A more complete understanding of the human mental model is necessary, which will require an even greater multidisciplinary approach and cooperation between diverse relevant researchers.

This study included some background terminologies such as Human–Computer Interaction (HCI), User Interface, User Interface Design, Usability, Universal Usability (Inclusive Design, Universal Design) and provided their scholastic definitions with their brief background. Next the literature survey focused on the foundation works laid out by the pioneering researchers in the knowledge domain. It described some of the early projects run by them and the problems they faced to initiate and introduce universal usability through plasticity of user interface design.

Description of the basic types of plasticity or adaptation: adaptive and adaptable were given. The differences and relationship among them as well as the issues related to adaptation were also discussed. In addition to that, these represent the three major components of adaptivity, viz.: Afferential Component (information about the user), Inferential Component (how the interface is modified) and Efferential Component (determines the need for the interface change based on the afferential and inferential components).

An historical overview of the path towards adaptation has been given, starting from the first adaptive project. It has focused on the early projects and how these overcame the obstacles faced and helped to pave the way towards universal usability through dynamic techniques (plasticity).

This study then focused on the different fields where universal usability has been achieved with the help of plasticity of interface design. These include: Contact Centre Agents (CCA), Large Scale Software Projects, Industrial Control Applications, Airborne Crew Stations, Adaptive Web Sites, Adaptive Hypermedia, Mental Workload Dynamics, Generation of Web Services and Construction of Query Forms, Adaptive Vehicular Communication System, Multimedia Courseware, Conversational Speech Processing, multimodal–multisensor Interfaces, especially Multimodality in Mobile Environment, Virtual Environments (especially three-dimensional navigation) using Rough Sets Theory, Spatial Applications and Adaptive Personalisation for services offered by mobile and wireless devices. Design and development tools to aid adaptation, such as the use of the Visual Language Generator (VLG), Adaptive GUI design tool architecture and Abstract User Interface Toolkit (AUIT) have been compared.

Various approaches to achieving universal usability through plasticity have been investigated and included: Runtime Model-based, use of Fuzzy Logic, Cognitive Modelling, Agile Adaptive User Interface Model, "System-controlled Adaptive Menus", "User-controlled Adaptable Menus" and finally the use of Knowledge Bases.

To investigate user personalisation, User Characteristics parameters such as: Personality Traits, Vocational Interests, Perceptual Style, Vocational Aptitude, Computer Experience and Application Experience, Psychological, Physical and Cognitive Human Characteristics, users' background, goals and competency, cultural-linguistic heritage and the application of graphics in multilingual websites have all been studied in depth.

Finally, the performance evaluation of plasticity of user interface has been conducted. Despite some negative aspects of plasticity of user interface design, it can be firmly concluded that the use of dynamic techniques can enhance the user experience to a much greater extent than more basic approaches. This also leads to the successful goal of achieving universal usability. Nonetheless optimisation of the usability parameter trade-offs needs attention.

It is noteworthy that most of the work reviewed has originated from a limited range of cultural perspectives. The present authors' experience indicates a need for much greater attention to this aspect as cross-cultural usability has not been the primary focus of universal usability research thus far. To make an interface simultaneously usable for users from a diverse range of cultural backgrounds will require a very large amount of adaptation, but the powerful principles of plasticity of user interface design hold the future promise of an optimum tool to achieve cross-cultural usability.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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